

# Restaurant Automation SaaS - API Documentation

## 1. Purpose

This document is the copy-paste reference for the backend API surface of the Restaurant Automation SaaS platform. It aligns with the role-wise PRD and the role-wise Postman collection so the backend, frontend, and testing flow stay consistent.

Source of truth: - restaurant\_automation\_final\_prd.md - restaurant\_automation\_rolewise\_postman\_and\_p

The platform uses a QR-session-first customer flow and JWT-based staff/admin flow: - Customers use temporary table sessions via **x-session-token** - Staff, restaurant admins, and super admin use JWT auth via **Authorization: Bearer <token>**

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## 2. API Standards

### Base URL

All HTTP APIs are versioned under:

/api/v1

### Standard success response

```
{
  "success": true,
  "data": {}
}
```

### Standard error response

```
{
  "success": false,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Validation failed"
  }
}
```

### Common error codes

- VALIDATION\_ERROR
- INVALID\_REQUEST
- UNAUTHORIZED

- TOKEN\_EXPIRED
- TOKEN\_INVALID
- FORBIDDEN
- NOT\_FOUND
- CONFLICT
- RATE\_LIMIT\_EXCEEDED
- TABLE\_SESSION\_EXPIRED
- INTERNAL\_ERROR

### Query param conventions

- Use path params for a single resource, for example `/items/:id`
- Use query params for filtering, sorting, pagination, and dashboards, for example:
  - `?page=1&limit=20`
  - `?status=READY`
  - `?search=burger`
  - `?from=2026-01-01&to=2026-01-31`

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## 3. Role-wise Postman Collection Structure

Restaurant Automation API

- Auth APIs
- Public APIs
- Customer APIs
- Service Staff APIs
- Kitchen APIs
- Cleaning APIs
- Restaurant Admin APIs
- Super Admin APIs
- Shared Utility APIs
- Webhook APIs
- Socket Testing APIs
- Health & Monitoring APIs

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## 4. Authentication Model

### 4.1 Staff / Admin / Kitchen / Cleaning

Use JWT auth: - Authorization: Bearer <accessToken>

### 4.2 Customer dine-in flow

Use QR/table sessions: - x-session-token: <sessionToken>

### 4.3 Optional customer account flow

Customer auth can exist for future loyalty, reservation history, and saved preferences, but dine-in ordering is session-first.

## 5. Auth APIs

| Method | Path                      | Purpose                                      |
|--------|---------------------------|----------------------------------------------|
| POST   | /auth/register            | Create a user account                        |
| POST   | /auth/login               | Login with password or supported credentials |
| POST   | /auth/request-otp         | Request OTP                                  |
| POST   | /auth/verify-otp          | Verify OTP                                   |
| POST   | /auth/refresh             | Refresh access token                         |
| POST   | /auth/logout              | Logout and clear session state               |
| POST   | /auth/forgot-password     | Start password recovery                      |
| POST   | /auth/reset-password      | Complete password reset                      |
| GET    | /auth/me                  | Get current authenticated user               |
| GET    | /auth/sessions            | List active sessions                         |
| DELETE | /auth/sessions/:sessionId | Revoke one session                           |

### Typical register body

```
{
  "name": "Om",
  "email": "om@example.com",
  "mobile": "9999999999",
  "password": "StrongPassword123"
}
```

## 6. Public APIs

| Method | Path                              | Purpose                        |
|--------|-----------------------------------|--------------------------------|
| GET    | /public/restaurants/:slug         | Public restaurant details      |
| GET    | /public/menu/:restaurantId        | Public menu for a restaurant   |
| POST   | /public/table-session/validate    | Validate QR table session      |
| POST   | /public/table-session/create      | Create table session           |
| GET    | /public/reservations/availability | Check reservation availability |

| Method | Path               | Purpose           |
|--------|--------------------|-------------------|
| POST   | /public/queue/join | Join public queue |

### Public menu query examples

GET /public/menu/:restaurantId?veg=true  
 GET /public/menu/:restaurantId?category=pizza  
 GET /public/menu/:restaurantId?spicy=false  
 GET /public/menu/:restaurantId?search=burger

## 7. Customer APIs

### 7.1 Customer session

| Method | Path                     | Purpose                   |
|--------|--------------------------|---------------------------|
| GET    | /customer/session        | Get current table session |
| PATCH  | /customer/session/extend | Extend session            |
| POST   | /customer/session/end    | End session               |

### 7.2 Customer menu

| Method | Path                      | Purpose              |
|--------|---------------------------|----------------------|
| GET    | /customer/menu/categories | Get menu categories  |
| GET    | /customer/menu/items      | Get menu items       |
| GET    | /customer/menu/items/:id  | Get single menu item |

### Menu query examples

GET /customer/menu/items?veg=true  
 GET /customer/menu/items?category=pizza  
 GET /customer/menu/items?available=true  
 GET /customer/menu/items?popular=true  
 GET /customer/menu/items?recommended=true  
 GET /customer/menu/items?search=pasta  
 GET /customer/menu/items?priceMin=100&priceMax=500  
 GET /customer/menu/items?sortBy=price

### 7.3 Customer cart

| Method | Path                         | Purpose          |
|--------|------------------------------|------------------|
| GET    | /customer/cart               | Get cart         |
| POST   | /customer/cart/items         | Add item to cart |
| PATCH  | /customer/cart/items/:itemId | Update cart item |
| DELETE | /customer/cart/items/:itemId | Remove cart item |
| DELETE | /customer/cart               | Clear cart       |

#### 7.4 Customer orders

| Method | Path                         | Purpose          |
|--------|------------------------------|------------------|
| POST   | /customer/orders             | Place order      |
| GET    | /customer/orders             | List orders      |
| GET    | /customer/orders/:id         | Get single order |
| POST   | /customer/orders/:id/reorder | Reorder          |
| POST   | /customer/orders/:id/cancel  | Cancel order     |

#### Order query examples

GET /customer/orders?status=PREPARING

GET /customer/orders?page=1&limit=10

#### 7.5 Customer assistance

| Method | Path                        | Purpose            |
|--------|-----------------------------|--------------------|
| POST   | /customer/requests/waiter   | Call waiter        |
| POST   | /customer/requests/water    | Request water      |
| POST   | /customer/requests/cutlery  | Request cutlery    |
| POST   | /customer/requests/cleaning | Request cleaning   |
| POST   | /customer/requests/help     | Request assistance |

#### 7.6 Customer billing

| Method | Path                            | Purpose            |
|--------|---------------------------------|--------------------|
| GET    | /customer/bill                  | Get live bill      |
| POST   | /customer/bill/request          | Request final bill |
| POST   | /customer/bill/coupon           | Apply coupon       |
| DELETE | /customer/bill/coupon/:couponId | Remove coupon      |

## 7.7 Customer payments

| Method | Path                                 | Purpose        |
|--------|--------------------------------------|----------------|
| POST   | /customer/payments/create            | Create payment |
| POST   | /customer/payments/verify            | Verify payment |
| GET    | /customer/payments/:paymentId/status | Payment status |

## 7.8 Customer feedback

| Method | Path               | Purpose          |
|--------|--------------------|------------------|
| POST   | /customer/feedback | Submit feedback  |
| GET    | /customer/feedback | Feedback history |

## 7.9 Customer loyalty

| Method | Path                         | Purpose           |
|--------|------------------------------|-------------------|
| GET    | /customer/loyalty            | Loyalty wallet    |
| GET    | /customer/offers             | Available offers  |
| GET    | /customer/offers/eligibility | Offer eligibility |

## 8. Service Staff APIs

| Method | Path                             | Purpose               |
|--------|----------------------------------|-----------------------|
| GET    | /staff/tables                    | Table dashboard       |
| GET    | /staff/tables/:id                | Table details         |
| PATCH  | /staff/tables/:id/assign         | Assign table          |
| PATCH  | /staff/tables/:id/reserve        | Mark reserved         |
| PATCH  | /staff/tables/:id/occupy         | Mark occupied         |
| GET    | /staff/queue                     | Queue dashboard       |
| GET    | /staff/queue/:id                 | Queue details         |
| PATCH  | /staff/queue/:id/priority        | Update queue priority |
| GET    | /staff/reservations              | Reservation list      |
| GET    | /staff/reservations/:id          | Reservation details   |
| PATCH  | /staff/reservations/:id/check-in | Check in reservation  |
| GET    | /staff/orders/ready              | Food pickup queue     |
| PATCH  | /staff/orders/:id/pick           | Pick order            |
| PATCH  | /staff/orders/:id/serve          | Mark served           |
| GET    | /staff/requests                  | Staff requests        |

| Method | Path                         | Purpose          |
|--------|------------------------------|------------------|
| PATCH  | /staff/requests/:id/accept   | Accept request   |
| PATCH  | /staff/requests/:id/complete | Complete request |
| POST   | /staff/issues/escalate       | Escalate issue   |

### Staff table query examples

GET /staff/tables?status=AVAILABLE

GET /staff/tables?floor=1

GET /staff/tables?section=VIP

## 9. Kitchen APIs

### Kitchen dashboard and order flow

| Method | Path                       | Purpose               |
|--------|----------------------------|-----------------------|
| GET    | /kitchen/dashboard         | Kitchen dashboard     |
| GET    | /kitchen/orders            | Kitchen orders list   |
| GET    | /kitchen/orders/:id        | Kitchen order details |
| PATCH  | /kitchen/orders/:id/accept | Accept order          |
| PATCH  | /kitchen/orders/:id/start  | Start cooking         |
| PATCH  | /kitchen/orders/:id/ready  | Mark ready            |
| PATCH  | /kitchen/orders/:id/delay  | Delay order           |
| PATCH  | /kitchen/orders/:id/reject | Reject order          |

### Kitchen query examples

GET /kitchen/orders?status=PREPARING

GET /kitchen/orders?priority=HIGH

GET /kitchen/orders?table=12

GET /kitchen/orders?batch=true

### Kitchen batching and metrics

| Method | Path                 | Purpose                  |
|--------|----------------------|--------------------------|
| GET    | /kitchen/batches     | List batches             |
| GET    | /kitchen/batches/:id | Batch details            |
| POST   | /kitchen/batches     | Create batch             |
| PATCH  | /kitchen/batches/:id | Update batch             |
| GET    | /kitchen/load        | Kitchen load metrics     |
| GET    | /kitchen/performance | Chef performance metrics |

## 10. Cleaning APIs

| Method | Path                         | Purpose               |
|--------|------------------------------|-----------------------|
| GET    | /cleaning/tasks              | Cleaning task list    |
| GET    | /cleaning/tasks/:id          | Cleaning task details |
| PATCH  | /cleaning/tasks/:id/start    | Start cleaning        |
| PATCH  | /cleaning/tasks/:id/complete | Complete cleaning     |
| PATCH  | /cleaning/tasks/:id/verify   | Verify cleaning       |

### Cleaning query examples

GET /cleaning/tasks?status=PENDING

GET /cleaning/tasks?priority=HIGH

## 11. Restaurant Admin APIs

### 11.1 Restaurant settings

| Method | Path                       | Purpose             |
|--------|----------------------------|---------------------|
| GET    | /admin/restaurant/overview | Restaurant overview |
| GET    | /admin/restaurant/settings | Restaurant settings |
| PATCH  | /admin/restaurant/settings | Update settings     |

### 11.2 Tables

| Method | Path                 | Purpose            |
|--------|----------------------|--------------------|
| POST   | /admin/tables        | Create table       |
| GET    | /admin/tables        | List tables        |
| GET    | /admin/tables/:id    | Table details      |
| PATCH  | /admin/tables/:id    | Update table       |
| DELETE | /admin/tables/:id    | Delete table       |
| POST   | /admin/tables/bulk   | Bulk create tables |
| POST   | /admin/tables/:id/qr | Generate table QR  |
| GET    | /admin/tables/:id/qr | Download table QR  |

### 11.3 Menu

| Method | Path                   | Purpose         |
|--------|------------------------|-----------------|
| POST   | /admin/menu/categories | Create category |

| Method | Path                               | Purpose             |
|--------|------------------------------------|---------------------|
| GET    | /admin/menu/categories             | List categories     |
| POST   | /admin/menu/items                  | Create menu item    |
| GET    | /admin/menu/items                  | List menu items     |
| GET    | /admin/menu/items/:id              | Menu item details   |
| PATCH  | /admin/menu/items/:id              | Update menu item    |
| DELETE | /admin/menu/items/:id              | Delete menu item    |
| PATCH  | /admin/menu/items/:id/availability | Toggle availability |
| POST   | /admin/menu/items/:id/image        | Upload menu image   |

### Admin menu query examples

GET /admin/menu/items?available=true  
 GET /admin/menu/items?category=starter  
 GET /admin/menu/items?search=burger

### 11.4 Staff management

| Method | Path                     | Purpose             |
|--------|--------------------------|---------------------|
| POST   | /admin/staff             | Create staff        |
| GET    | /admin/staff             | List staff          |
| GET    | /admin/staff/:id         | Staff details       |
| PATCH  | /admin/staff/:id         | Update staff        |
| DELETE | /admin/staff/:id         | Delete staff        |
| POST   | /admin/staff/shifts      | Shift assignment    |
| GET    | /admin/staff/attendance  | Attendance list     |
| GET    | /admin/staff/performance | Performance metrics |

### 11.5 Offers and loyalty

| Method | Path                 | Purpose             |
|--------|----------------------|---------------------|
| POST   | /admin/offers        | Create offer        |
| GET    | /admin/offers        | List offers         |
| GET    | /admin/offers/:id    | Offer details       |
| PATCH  | /admin/offers/:id    | Update offer        |
| DELETE | /admin/offers/:id    | Delete offer        |
| GET    | /admin/loyalty/rules | Loyalty rules       |
| POST   | /admin/loyalty/rules | Create loyalty rule |

### 11.6 Billing reports

| Method | Path                     | Purpose          |
|--------|--------------------------|------------------|
| GET    | /admin/billing/revenue   | Daily revenue    |
| GET    | /admin/billing/tax       | Tax reports      |
| GET    | /admin/billing/orders    | Order reports    |
| GET    | /admin/billing/discounts | Discount reports |
| GET    | /admin/billing/payments  | Payment reports  |

### 11.7 Analytics

| Method | Path                                | Purpose            |
|--------|-------------------------------------|--------------------|
| GET    | /admin/analytics/revenue            | Revenue analytics  |
| GET    | /admin/analytics/peak-hours         | Peak hours         |
| GET    | /admin/analytics/repeat-customers   | Repeat customers   |
| GET    | /admin/analytics/kitchen            | Kitchen analytics  |
| GET    | /admin/analytics/table-utilization  | Table utilization  |
| GET    | /admin/analytics/customer-retention | Customer retention |

#### Revenue analytics query examples

GET /admin/analytics/revenue?from=2026-01-01&to=2026-01-31

GET /admin/analytics/revenue?groupBy=day

GET /admin/analytics/revenue?groupBy=month

### 11.8 Inventory

| Method | Path                    | Purpose           |
|--------|-------------------------|-------------------|
| GET    | /admin/inventory        | Ingredient list   |
| POST   | /admin/inventory        | Add ingredient    |
| PATCH  | /admin/inventory/:id    | Update ingredient |
| GET    | /admin/inventory/alerts | Stock alerts      |

## 12. Super Admin APIs

### 12.1 Platform and tenant management

| Method | Path                           | Purpose            |
|--------|--------------------------------|--------------------|
| GET    | /super-admin/platform/overview | Platform overview  |
| GET    | /super-admin/restaurants       | Restaurant list    |
| GET    | /super-admin/restaurants/:id   | Restaurant details |

| Method | Path                                 | Purpose            |
|--------|--------------------------------------|--------------------|
| PATCH  | /super-admin/restaurants/:id/approve | Approve restaurant |
| PATCH  | /super-admin/restaurants/:id/suspend | Suspend restaurant |
| DELETE | /super-admin/restaurants/:id         | Delete restaurant  |

### Restaurant list query examples

GET /super-admin/restaurants?status=ACTIVE

GET /super-admin/restaurants?plan=PRO

GET /super-admin/restaurants?search=pizza

## 12.2 Plans, analytics, and platform controls

| Method | Path                           | Purpose                    |
|--------|--------------------------------|----------------------------|
| POST   | /super-admin/plans             | Create subscription plan   |
| GET    | /super-admin/plans             | List plans                 |
| PATCH  | /super-admin/plans/:id         | Update plan                |
| GET    | /super-admin/analytics/revenue | Platform revenue analytics |
| GET    | /super-admin/analytics/tenants | Active tenant analytics    |
| GET    | /super-admin/system/monitoring | System monitoring          |
| GET    | /super-admin/audit-logs        | Audit logs                 |
| GET    | /super-admin/feature-flags     | Feature flags              |
| PATCH  | /super-admin/feature-flags/:id | Update feature flag        |

### Audit log query examples

GET /super-admin/audit-logs?actorId=123

GET /super-admin/audit-logs?action=DELETE

GET /super-admin/audit-logs?from=2026-01-01&to=2026-01-10

## 13. Shared Utility APIs

| Method | Path                    | Purpose                     |
|--------|-------------------------|-----------------------------|
| GET    | /notifications          | List notifications          |
| PATCH  | /notifications/:id/read | Mark notification read      |
| PATCH  | /notifications/read-all | Mark all notifications read |
| POST   | /uploads                | File upload                 |
| GET    | /search?q=pizza         | Global search               |

## 14. Webhook APIs

Webhook routes are reserved for future third-party integrations such as: - payment gateway callbacks - SMS callbacks - email delivery callbacks - external automation callbacks

If implemented later, keep webhook verification strict and signature-based.

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## 15. Socket Testing APIs

Socket testing is used to verify live events for: - order updates - kitchen queue updates - billing updates - reservation updates - notifications - table state updates - staff request updates

Recommended event groups: - order:new - order:status-updated - kitchen:batch-created - kitchen:batch-updated - table:status-updated - notification:new - session:created - session:expired

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## 16. Health and Monitoring APIs

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| Method | Path     | Purpose         |
|--------|----------|-----------------|
| GET    | /health  | Liveness check  |
| GET    | /ready   | Readiness check |
| GET    | /version | Version check   |

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## 17. QR Session Security Notes

- QR creates a temporary table session
- Session expires after dining/payment
- Old URL cannot reopen the table
- Reusing an expired session must return a clear error

Example:

```
{
  "success": false,
  "error": {
    "code": "TABLE_SESSION_EXPIRED",
    "message": "Please scan another QR code."
  }
}
```

---

## 18. Postman Environment Variables

Use these in the workspace collection:

```
baseUrl
accessToken
refreshToken
restaurantId
tableId
sessionId
orderId
menuItemId
reservationId
staffId
customerId
paymentId
billId
offerId
batchId
notificationId
```

---

## 19. Recommended Request Flow for Testing

1. Auth
  2. Public QR session create/validate
  3. Customer menu browse
  4. Cart add/update/remove
  5. Order place and order status tracking
  6. Kitchen accept/start/ready
  7. Staff pick/serve
  8. Billing request and coupon flow
  9. Payment create/verify
  10. Session close
  11. Cleaning task verification
  12. Admin analytics and inventory checks
  13. Super admin tenant checks
  14. Notification and socket event checks
- 

## 20. Workspace and Run Notes

This project uses an npm workspace setup. Recommended commands:

```
npm install
npm run dev --workspace backend
```

```
npm run dev --workspace frontend
npm run typecheck
npm run verify:phase1 --workspace backend
docker-compose up --build
```

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## 21. Summary

This API documentation must stay aligned with: - backend route ownership - PRD expectations - session-first customer UX - JWT-based staff/admin auth - role-wise Postman collection structure

If an endpoint is changed, the documentation and the Postman collection must be updated at the same time.